

HireAI

AI Recruitment Platform

Technical Documentation and User Manual

Also referred to as the Role-Based Dashboard System

Infosys Springboard Virtual Internship 7.0

Project: HireAI — AI Recruitment Platform

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Document Control

Table DC-1. Revision History

Version	Date	Description	Prepared by
1.0	15 Sep 2026	Initial release of the technical documentation and user manual.	Hemanth M

This document describes the HireAI project as implemented in the repository referenced above. Where the underlying implementation has been audited at different points in development, this document uses the most recent audit snapshot available at the time of writing (see Section 11) as the reference for current status.

This is a project/academic technical document, not a specification for a certified or finished commercial product. Features, ports, and limitations described here reflect the reviewed state of the codebase and may change as development continues.

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Section 1 — Introduction

1.1 Project Overview

HireAI is a full-stack web platform built to support the recruitment and candidate-evaluation process. It combines role-based access control, resume analysis, AI-assisted mock interviews, behavioral analysis, structured interview scoring, AI-generated feedback, candidate and recruiter analytics, notifications, scheduling, and reporting within a single web application. The system is also referred to internally as the Role-Based Dashboard System, reflecting its organization around three distinct user roles.

The project is maintained in the repository *springboardmentor98765a-tech/AI_Mock-Interview-and-Candidate-Assessment-Platform*, with the primary branch currently documented being *Hemanth_M*.

1.2 Problem Statement

Recruitment workflows commonly span several disconnected tools: resume screening, interview scheduling, interview evaluation, and candidate comparison are often handled separately, and scoring between interviewers can be inconsistent. HireAI addresses this by bringing these stages into one platform and applying a consistent, weighted scoring framework (Module 7) together with structured, evidence-based feedback to every mock interview conducted on the system.

1.3 Objectives

- Provide a single platform covering the recruitment workflow from account creation through candidate comparison and shortlisting.
- Apply a consistent, weighted scoring model to every mock interview through the Module 7 scoring engine.
- Give candidates measurable feedback and analytics that can guide their preparation.
- Give recruiters ranking, comparison, and shortlisting tools grounded in stored interview data.
- Give administrators visibility into platform usage and AI service activity.

1.4 Scope

HireAI covers candidate registration and authentication, resume upload and analysis, AI-assisted mock interviews (including speech and behavioral evaluation where the corresponding services are active), automated scoring and feedback, candidate analytics, recruiter ranking and comparison, and administrative monitoring. The platform is intended to provide assessments, analytics, and decision-support recommendations. It does not make legally binding hiring decisions, and it does not replace human recruiters; final hiring decisions remain with the organization using the platform.

1.5 Intended Users

- Candidates preparing for and undertaking mock interviews.
- Recruiters reviewing, ranking, and comparing candidates.
- Administrators managing platform users and monitoring system and AI service activity.

Section 2 — System Overview

2.1 Platform Overview

HireAI organizes its functionality around three roles — Candidate, Recruiter, and Administrator — each with a dedicated dashboard. The platform aims to bring several stages of the recruitment workflow into one application: account creation and authentication, resume upload and analysis, AI-assisted role recommendations, mock interview setup and delivery, speech and behavioral evaluation, weighted interview scoring, AI feedback, candidate performance analytics, recruiter candidate ranking and comparison, shortlist insights, reporting, notifications and scheduled interviews, and administrative monitoring.

2.2 Roles

Table 2. User Roles

Role	Main Responsibilities	Key Dashboard
Candidate	Registers, uploads a resume, takes mock interviews, reviews scores, feedback and analytics, downloads available reports.	Candidate Dashboard
Recruiter	Reviews candidates, ranks and compares candidates, reviews shortlisting insights, accesses available reports.	Recruiter Dashboard
Administrator	Manages users and recruiters, monitors interview and AI service activity, reviews platform usage analytics.	Admin Dashboard

2.3 End-to-End Workflow

Candidate Workflow

- Open HireAI and register or log in, optionally using Google or GitHub OAuth where configured.
- Reach the candidate dashboard.
- Upload a resume; resume analysis extracts skills and technologies and can produce an ATS-related result.
- Start a mock interview, selecting role, interview type, difficulty, and question count as supported by the interface.
- Answer questions through the live interview interface; speech processing can provide transcription and acoustic information, and behavioral/CV analysis can provide engagement- and confidence-related indicators when the CV service is active.
- Receive a Module 7 score across four categories and an overall weighted score, together with structured AI feedback.

- Review interview history, competency analytics, trends, weak areas, and improvement information, and download available reports.

Recruiter Workflow

- Log in and open the recruiter dashboard.
- View candidate and performance information based on persisted platform data.
- View candidate rankings; search, sort, paginate, and export ranking information where supported.
- Inspect candidate details, including interview results and available analysis.
- Compare multiple candidates side-by-side (up to four in the implemented comparison workflow).
- View skill/competency analytics and interview activity/performance trends.
- View deterministic shortlisting insights — category, reasons, strengths, and risks — where available, and access reports.

Administrator Workflow

- Sign in and open the admin dashboard.
- Access platform-level management and monitoring features supported by the current implementation.
- Manage users and recruiters, and monitor interview activity.
- Inspect AI usage telemetry and system health/activity metrics.
- View platform usage analytics and access administrative reporting where supported.

2.4 Major Features

- Role-based access control across Candidate, Recruiter, and Administrator roles.
- Resume upload, parsing, and ATS-related analysis.
- AI-assisted mock interviews with speech and behavioral evaluation.
- Module 7 weighted interview scoring and structured AI feedback.
- Candidate analytics: history, competency breakdown, trends, and weak-area identification.
- Recruiter ranking, comparison, and deterministic shortlisting insight tools.
- Administrative monitoring of users, interview activity, AI usage, and system health.
- Notifications, interview scheduling, and reporting.

Section 3 — Technology Stack

3.1 Frontend

The frontend is a React single-page application built with Vite.

- React 18 (react ^18.2.0, react-dom ^18.2.0)
- React Router 6 (react-router-dom ^6.20.0) for client-side routing
- Vite 5 (^5.0.0) as the build tool and development server
- Recharts (^3.10.1) for charts and data visualization
- Framer Motion (^12.42.2) for UI animation
- Lucide React (^1.27.0) for icons
- A project-maintained plain CSS stylesheet system

3.2 Backend

The backend is a Node.js application built on Express.js.

- Node.js and Express.js as the application server and REST API layer
- pg as the PostgreSQL driver
- Passport.js, with passport-google-oauth20 and passport-github2 strategies
- jsonwebtoken for JWT-based authentication
- bcryptjs for password hashing
- multer for file upload handling (for example, resumes)
- nodemailer for email delivery
- helmet, express-rate-limit, and express-validator for security and input validation
- pdfkit and csv-stringify for server-side report generation
- @google/genai for Gemini LLM integration
- dotenv for environment configuration

Backend package scripts: npm start runs node server.js; npm run dev runs nodemon server.js.

3.3 Database

PostgreSQL is the persistent application database. The project database is documented under the name ai_recruitment. Tables are initialized and migrated through backend configuration code, queries use parameterized statements, and JSONB is used for structured interview and evaluation data.

3.4 Authentication

- Traditional email/password authentication with bcryptjs password hashing

- JWT-based authentication for API access
- Google OAuth 2.0 via Passport.js
- GitHub OAuth via Passport.js
- Role-based access control

3.5 AI Components

HireAI is not a single AI model but an AI-enabled pipeline made up of several components: an LLM text-generation provider, a speech-to-text service, a text-to-speech service, a computer-vision behavioral analysis service, and resume-analysis logic in the backend. Each is described in Section 4.5 and Section 5.

3.6 Supporting Libraries

Security and supporting middleware used across the backend includes Helmet, CORS configuration, express-rate-limit, and express-validator, alongside pdfkit and csv-stringify for reporting and nodemailer for email delivery, as listed above.

Table 1. Technology Stack

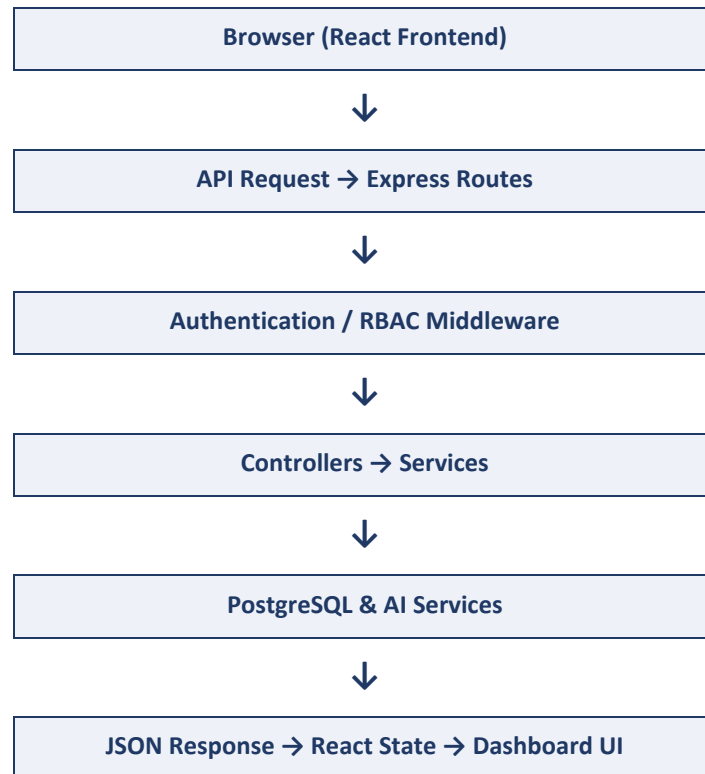
Layer	Technology	Purpose
Frontend	React 18, React Router 6, Vite 5	UI rendering and client-side routing
Frontend	Recharts	Data visualization and charts
Frontend	Framer Motion	UI animation
Frontend	Lucide React	Icon set
Backend	Node.js, Express.js	Application server and REST API
Backend	pg	PostgreSQL driver
Backend	Passport.js (Google / GitHub strategies)	OAuth authentication
Backend	jsonwebtoken, bcryptjs	JWT authentication and password hashing
Backend	multer, nodemailer	File uploads and email delivery
Backend	helmet, express-rate-limit, express-validator	Security middleware and input validation
Backend	pdfkit, csv-stringify	Server-side report generation
Backend	@google/genai	Gemini LLM integration

Layer	Technology	Purpose
Database	PostgreSQL	Persistent application data store
AI Services	Faster-Whisper, Kokoro, PyTorch / OpenCV / ResNet-18 / YuNet	Speech-to-text, text-to-speech, and behavioral (CV) analysis

Section 4 — System Architecture

4.1 High-Level Architecture

At a high level, a request travels from the browser through the React frontend to the Express backend, through authentication and role-based access control middleware, into controllers and services, which interact with PostgreSQL and the AI services before returning a JSON response that updates the React application state and, in turn, the dashboard UI.



4.2 Frontend Architecture

The frontend root includes a `src/` directory organized into `context/` (React context providers), `services/` (API service functions), `pages/` (top-level route pages), and `components/` (reusable UI components). Documented frontend pages include `Login.jsx`, `Register.jsx`, `OAuthCallback.jsx`, `Settings.jsx`, `AdminDashboard.jsx`, `RecruiterDashboard.jsx`, `StudentDashboard.jsx`, `MockInterview.jsx`, and `DashboardLayout.jsx`.

4.3 Backend Architecture

The backend entry point is `backend/server.js`. It mounts the following route groups: `/api/auth`, `/api/resume`, `/api/interviews`, `/api/recordings`, `/api/interview`, `/api/stt`, `/api/cv`, `/api/analytics`, `/api/admin`, `/api/notifications`, `/api/schedules`, and `/api/reports`. The backend also connects to

PostgreSQL, initializes and migrates database structures, starts an in-process reminder scheduler after database initialization, exposes a /api/health endpoint, and uses centralized error handling.

The backend directory is organized into config/ (database and Passport/OAuth configuration), controllers/ (request-handling logic), routes/ (Express route definitions), middleware/ (authentication, RBAC, and request processing), models/ (data-access logic), services/ (business logic and integrations), and ai/ (the Python AI services).

4.4 Database Layer

PostgreSQL stores information related to users, interviews, interview questions and answers, recordings, resume analysis, scheduling, notifications, analytics and scoring data, CV analysis results, and report-related data. Implementation characteristics include parameterized SQL queries, JSONB for structured category and evaluation data, idempotent migration behavior for certain schema additions, transaction handling for critical interview-completion persistence, and database-backed analytics in the candidate, recruiter, and admin services.

4.5 AI Service Layer

The AI layer is composed of independent services rather than a single model.

Table 3. AI Services

Service	Technology	Purpose	Port
LLM	Gemini / Ollama-Qwen	Text generation and AI evaluation	Provider-dependent
STT	Faster-Whisper	Speech-to-text and speech analysis	8765
TTS	Kokoro	Text-to-speech	8766
CV	PyTorch + OpenCV + ResNet-18 + YuNet	Behavioral / video analysis	8767

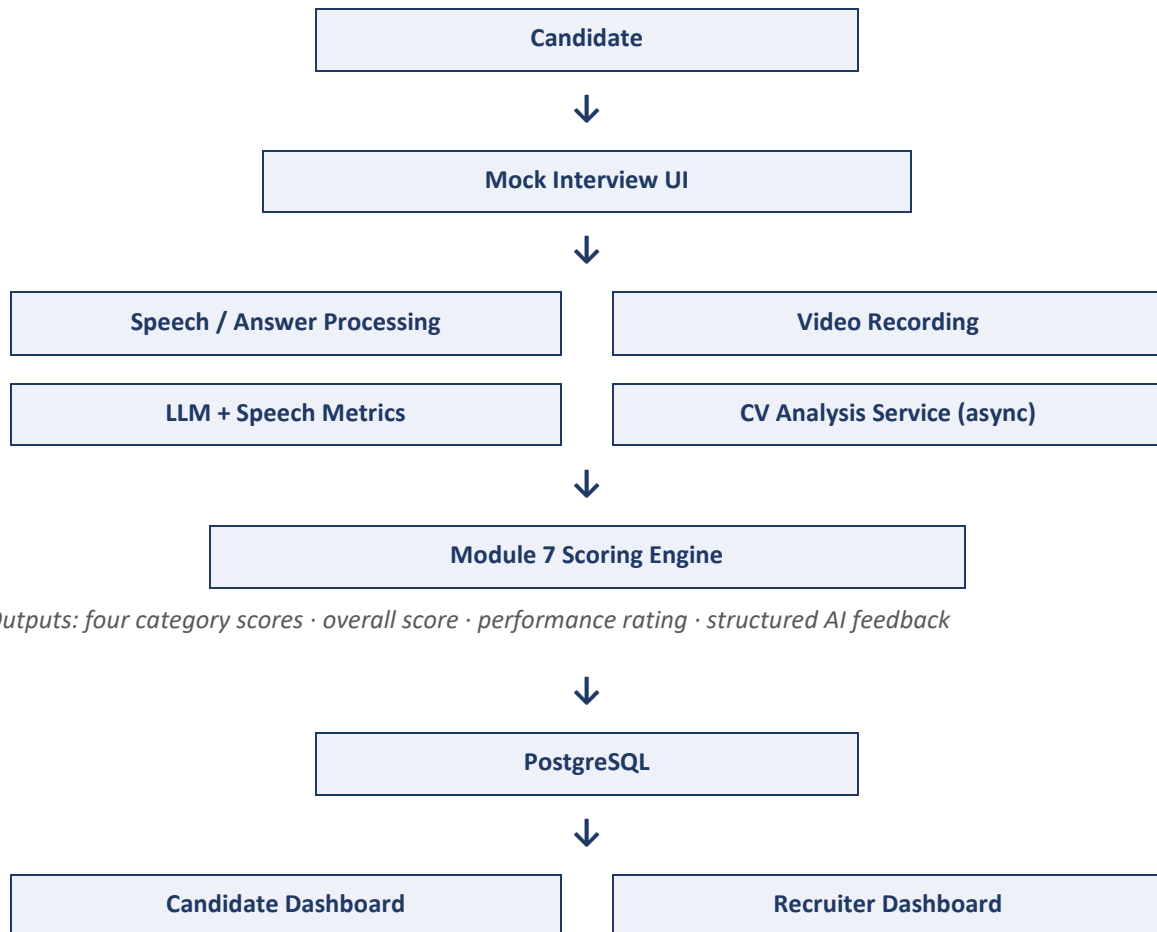
Local STT, TTS, and CV features depend on their corresponding Python services being installed and running; they are not purely browser-based features.

4.6 Service-to-Service Communication

The React frontend communicates with the Express backend over REST. The backend, in turn, communicates with the Gemini API (or a local Ollama/Qwen provider) for text generation, and with the STT, TTS, and CV Python services over internal HTTP on their respective ports. In the reviewed demonstration environment, the backend ran on port 5000 and the frontend development server on port 5173; PostgreSQL and the AI services remained on private ports (5432, 8765, 8766, 8767, and 11434 for Ollama where used), with only the backend and frontend ports exposed publicly through the Codespace forwarded-port layer.

4.7 Interview Data Flow

The following describes the sequence of processing for a single mock interview, from candidate input through to dashboard presentation.



Candidate Dashboard → Analytics / History / Weak Areas / Progress / Reports

Behavioral results from the CV analysis service can trigger an updated Module 7 score once CV analysis completes; the audited implementation includes asynchronous, background CV processing rather than having every interview wait synchronously for every service.

Section 5 — Functional Modules

5.1 Authentication

Candidates, recruiters, and administrators authenticate using email and password (hashed with bcryptjs) or, where configured, Google or GitHub OAuth through Passport.js. Successful authentication issues a JWT used for subsequent API access, and role-based access control determines which routes and dashboard features a given account can reach.

5.2 Resume Analysis

Resume files are uploaded through the backend, which parses resume content and stores analysis information in the database. The candidate dashboard exposes extracted skills, extracted technologies, and an ATS-related score, presented separately from interview-derived competency scores.

5.3 Role Recommendation

Following resume analysis, the platform can present AI-assisted role recommendations to the candidate as part of the mock-interview setup flow.

5.4 Mock Interview

A candidate configures a mock interview by selecting role, interview type, difficulty, and question count as supported by the interface. The system presents interview questions, and the candidate answers through the live interview interface, which includes timer, microphone, and camera controls alongside the behavioral-analysis panel where the CV service is active.

5.5 Speech Processing

The speech-to-text service (`backend/ai/stt_service.py`) uses Faster-Whisper, documented as configured with the Whisper “small” model, to transcribe candidate speech and provide speech-related measurements used in interview evaluation.

5.6 Computer Vision Analysis

The computer-vision service (`backend/ai/cv_service.py`) uses PyTorch, torchvision, OpenCV, a ResNet-18 trained model, and the YuNet ONNX face detector. Documented model files include `best_checkpoint.pt`, `face_detection_yunet_2023mar.onnx`, and `class_labels.json`. The pipeline produces behavioral and visual signals such as face visibility, a confidence-related indicator, an engagement estimate, attention-related metrics, an emotion-related estimate, and live compliance/warning information. These outputs are behavioral indicators only and are not presented as medical or psychological diagnoses.

5.7 AI Scoring

Every completed interview is scored by the Module 7 scoring engine across four weighted categories, producing an overall score on a 0–100 scale and a performance rating. The full scoring model, including the formula and category weights, is described in Section 6.

5.8 AI Feedback

Module 7 includes structured feedback across five categories: Strengths, Weaknesses, Suggestions, Practice Recommendations, and Learning Resources. Feedback is generated from the interview's measured and evaluated values and does not replace the scoring engine.

5.9 Candidate Analytics

The candidate dashboard presents interview history backed by database records; a per-interview detail view with question/answer information, transcript, speech analysis, recording playback, score breakdown, and feedback where available; competency analytics across the four Module 7 categories, including progress indicators and a radar chart; and performance trends visualizing chronological interview performance.

Weak-area identification uses deterministic historical score rules rather than a machine-learning probability model: a score threshold of 65 is used as an important low-score marker, repeated low performance contributes to risk classification, and early-versus-later historical performance is used to identify trend direction, producing a risk level, a reason, and a category-targeted recommendation. Improvement-progress logic compares earlier and later interview performance rather than projecting a future prediction.

5.10 Recruiter Analytics

The recruiter dashboard presents a candidate performance overview (candidate count, completed interview count, average AI score, score distribution, and activity information), candidate ranking based on real PostgreSQL data using SQL ranking logic (documented as `RANK() OVER (ORDER BY iv.score DESC NULLS LAST)`) with search, sort, pagination, and CSV export where available, candidate skill analytics comparing the top candidate against platform-level averages across the four Module 7 categories, and weekly interview activity/performance trend data aggregated over a 12-week period.

5.11 Admin Monitoring

The Module 10 audit reports the admin dashboard as substantially implemented with real backend/database integration, covering user and recruiter management (listing, role modification, account-status controls), interview activity monitoring (completed/in-progress/pending breakdowns, monthly activity, recent activity), AI performance monitoring (Gemini telemetry, call counts, usage counters, active-key information where supported), system activity and health (Node.js process information, memory/CPU-related metrics, uptime, database connection-pool information and

connectivity checks), and platform usage analytics (role distribution, monthly signups/activity aggregation).

5.12 Notifications and Scheduling

The backend includes routes and services for notifications, schedules, and reports, together with an in-process reminder scheduler started after PostgreSQL initialization. This supports scheduled interviews, reminder functionality, in-app notifications, and email delivery through Nodemailer where configured. External calendar integrations are not part of the current implementation.

5.13 Reports

Client-side text (TXT) report generation is available. Server-side PDF and CSV report functionality also exists using PDFKit and csv-stringify, but the latest code audit identified implementation defects in server-side candidate report data assembly and recruiter candidate-report wiring; this is treated as a current limitation (see Section 13) until repaired.

Section 6 — Scoring and AI Logic

6.1 Module 7

Module 7 is the canonical interview scoring layer used across the platform. It produces four category scores, a weighted overall score, a performance rating, and structured AI feedback for every completed interview.

6.2 Category Weights

Table 4. Module 7 Scoring

Category	Weight	Main Evidence
Communication	30%	Speech analysis combined with LLM evaluation
Confidence	25%	CV behavioral evidence combined with LLM evaluation
Technical Relevance	30%	Technical and problem-solving evaluation
Professionalism	15%	Grammar/professional communication, organization, time management, and interview etiquette

6.3 Overall Score Formula

The overall score is a weighted sum of the four category scores, kept on a 0–100 scale:

$$\text{Overall Score} = (\text{Communication} \times 0.30) + (\text{Confidence} \times 0.25) + (\text{Technical Relevance} \times 0.30) + (\text{Professionalism} \times 0.15)$$

6.4 Performance Rating

Table 5. Performance Ratings

Score Range	Rating
90–100	Excellent
75–89	Good
60–74	Average
40–59	Needs Improvement
0–39	Poor

6.5 Communication

The Communication category uses a blended source involving speech analysis and LLM evaluation.

6.6 Confidence

The Confidence category can incorporate CV behavioral evidence together with LLM evaluation.

6.7 Technical Relevance

The Technical Relevance category combines technical and problem-solving evaluation of the candidate's answers.

6.8 Professionalism

The Professionalism category incorporates grammar and professional communication, organization, time management, and interview etiquette/compliance indicators. The following tables describe the deterministic scoring bands used internally for time management and etiquette; they are part of the technical scoring model and are not exposed in this level of detail to end users.

Table 6a. Response Duration Scoring

Response Duration	Score
30–120 seconds	100
121–180 seconds	80
16–29 seconds	70
More than 180 seconds	60
15 seconds or less	40

Empty or zero-time answers are not treated as an ordinary measured duration.

Table 6b. Interview Etiquette / Compliance Scoring

Warning Count	Score
0	100
1	80
2	60
3	40
4 or more	20

6.9 Weak-Area Analysis

Weak-area identification is deterministic analytics logic rather than a machine-learning probability model. A score threshold of 65 is used as an important low-score marker, repeated low performance

contributes to risk classification, and early-versus-later historical performance is used to identify trend direction. The output can include a risk level, a reason, and a category-targeted recommendation.

6.10 Improvement Progress

Longitudinal improvement tracking compares a candidate's earlier and later interview performance and is presented in both the analytics service and the candidate dashboard. It reflects observed historical change rather than a predicted future outcome.

6.11 Shortlisting Insights

A deterministic shortlisting-insight layer categorizes candidates into hiring tiers and can provide a category, a reason, strengths, and risks for each candidate. This is presented to recruiters as decision-support insight and structured shortlist guidance rather than an autonomous hiring decision.

6.12 AI Feedback Generation

AI feedback is generated from the interview's measured and evaluated values across the five feedback categories described in Section 5.8 (Strengths, Weaknesses, Suggestions, Practice Recommendations, and Learning Resources). Feedback generation is evidence-based and does not replace or override the Module 7 scoring engine.

Section 7 — Database and API Design

7.1 Database Overview

PostgreSQL is the persistent application database (documented database name: ai_recruitment). It stores users, interviews, interview questions and answers, recordings, resume analysis, scheduling, notifications, analytics and scoring data, CV analysis results, and report-related data.

7.2 Data Persistence

- Parameterized SQL queries throughout the data-access layer.
- JSONB used for structured category and evaluation data.
- Idempotent migration behavior for at least certain schema additions.
- Transaction handling for critical interview-completion persistence.
- Database-backed analytics in the candidate, recruiter, and admin services.

7.3 API Organization

The backend mounts route groups by functional area from backend/server.js: authentication, resume, interviews, recordings, interview, STT, CV, analytics, admin, notifications, schedules, and reports, alongside a health-check endpoint.

7.4 Authentication Flow

A candidate, recruiter, or administrator authenticates with email and password or via Google/GitHub OAuth through Passport.js. On success, the backend issues a JWT, which the frontend attaches to subsequent API requests; role-based access control middleware then authorizes each request against the account's role before it reaches the corresponding controller.

7.5 Important API Categories

The table below groups the backend's routes by functional area rather than listing every individual endpoint; exact request and response payloads are not reproduced here unless the corresponding endpoint file has been directly inspected.

Table 6. Main API Groups

Area	Base Path	Purpose
Authentication	/api/auth	Registration, login, profile, password management, logout, Google/GitHub OAuth
Resume	/api/resume	Resume upload, parsing, analysis, status, and result retrieval

Area	Base Path	Purpose
Interviews	/api/interviews, /api/interview	Interview creation, question generation, completion, history, and detail retrieval
Recordings	/api/recordings	Recording upload, interview results, and detail retrieval
Speech-to-Text	/api/stt	Speech transcription and speech-related analysis
Computer Vision	/api/cv	Behavioral / video analysis
Analytics	/api/analytics	Candidate and recruiter analytics
Administration	/api/admin	User management and platform telemetry
Notifications	/api/notifications	In-app notification actions
Schedules	/api/schedules	Scheduled interview operations
Reports	/api/reports	Candidate, recruiter, and administrative reporting
Health	/api/health	Service health check

7.6 Error Handling

The backend applies centralized error handling around its route groups. As with the API category reference above, this document does not enumerate exact error payloads or status-code conventions beyond what is stated here, in keeping with the project's factuality requirements.

Section 8 — Security

8.1 Authentication

HireAI supports traditional email/password authentication alongside Google OAuth 2.0 and GitHub OAuth, both implemented through Passport.js strategies (passport-google-oauth20, passport-github2).

8.2 Authorization / RBAC

Role-based access control governs which routes and dashboard features a given account can reach, based on whether the account is a Candidate, Recruiter, or Administrator.

8.3 Password Security

Passwords are hashed using bcryptjs; plaintext passwords are not stored.

8.4 API Security

The backend applies Helmet for HTTP header hardening, CORS configuration to control cross-origin access, express-rate-limit to limit request rates, and express-validator for input validation on incoming requests.

8.5 Database Security

Database access uses parameterized SQL statements throughout, which mitigates SQL injection risk in the queries reviewed.

8.6 File Protection

Resumes and interview recordings are intended to be accessed through authenticated API routes rather than exposed as public static files.

8.7 Secret Management

Secrets — including database credentials, the JWT signing secret, Gemini API keys, and OAuth client credentials — are loaded from environment variables rather than being hard-coded. This document uses placeholders such as <DB_PASSWORD>, <GEMINI_API_KEY>, and <GOOGLE_CLIENT_ID> in place of any real values; see Appendix B.

The application applies common web-application security controls appropriate to its architecture and scale. It has not undergone formal third-party security certification and should not be presented as a certified secure system.

Section 9 — User Manual

9.1 Getting Started

- Open the HireAI application in a web browser.
- Select Register to create a new account, or Login if an account already exists.
- Optionally select Google or GitHub to sign in using an existing account, where this option is configured.
- After signing in, the application opens the dashboard for the account's role: Candidate, Recruiter, or Administrator.

9.2 Candidate Instructions

- Open the candidate dashboard after logging in.
- Open the resume section and upload a resume file to receive extracted skills, technologies, and an ATS-related result.
- Open the Mock Interview section.
- Choose the required role, interview type, difficulty, and question count.
- Start the interview and answer each question through the live interview interface; allow microphone and camera access if behavioral or speech analysis is offered.
- After the interview ends, review the overall score, the four category scores, and the AI feedback.
- Open the analytics tabs to review competency breakdown, performance trends, and weak-area information.
- Open interview history to review past interviews, and use the available report option to download a report.

9.3 Recruiter Instructions

- Log in and open the recruiter dashboard.
- Review the candidate performance overview for platform-wide statistics.
- Open the ranking view to see candidates ordered by score; use search, sort, and pagination as needed.
- Select a candidate to open their detail view, including interview results and available analysis.
- Select up to four candidates to open the side-by-side comparison view.
- Open the shortlisting insights area to review category, reasons, strengths, and risks for candidates.
- Use the reports section to access available candidate or recruiter reports.

9.4 Administrator Instructions

- Log in and open the admin dashboard.
- Open user management to review accounts, adjust roles, or change account status.
- Open interview activity monitoring to review completed, in-progress, and pending interviews.
- Open AI monitoring to review usage counters and telemetry for the AI services.
- Open system health to review process, memory, and database connectivity information.
- Open platform usage analytics for role distribution and signup/activity trends.

9.5 Reports

Candidates can generate a client-side text (TXT) report from their interview history. Server-generated PDF and CSV reports also exist in the platform, but current implementation defects in server-side report assembly mean these should be treated as a limited or in-progress feature rather than a fully verified export path (see Section 13).

9.6 Notifications

In-app notifications and scheduled-interview reminders are available where visible in the interface. Email delivery for notifications uses Nodemailer where email is configured.

9.7 Settings

The Settings page allows an account holder to review and update profile and preference controls.

9.8 Common Issues

- If the behavioral analysis panel does not appear during an interview, the CV service may not be running; this is a separate Python service rather than a browser-only feature.
- If speech transcription is unavailable, confirm that the STT service is installed and running.
- If a report download does not complete as expected, use the client-side text report as a fallback while the known server-side report defects are addressed.
- If Google or GitHub sign-in fails, confirm that OAuth is configured for the environment being used.
- Camera and microphone browser permissions must be granted for the live interview's speech and behavioral features to function.

Section 10 — Screenshot-Based Walkthrough

This section reserves a labeled placeholder for each screen referenced elsewhere in this document. Each placeholder should be replaced with an actual screenshot of the running application; see Appendix D for the full screenshot index.

Getting Started

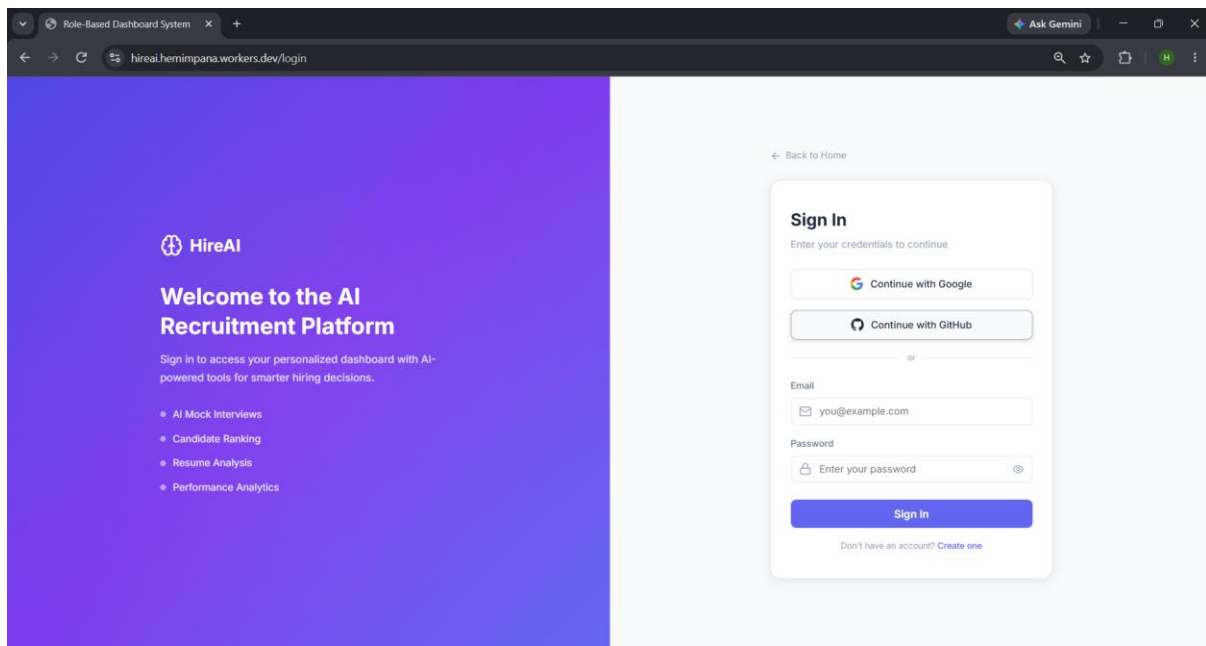


Figure 1. HireAI login screen.

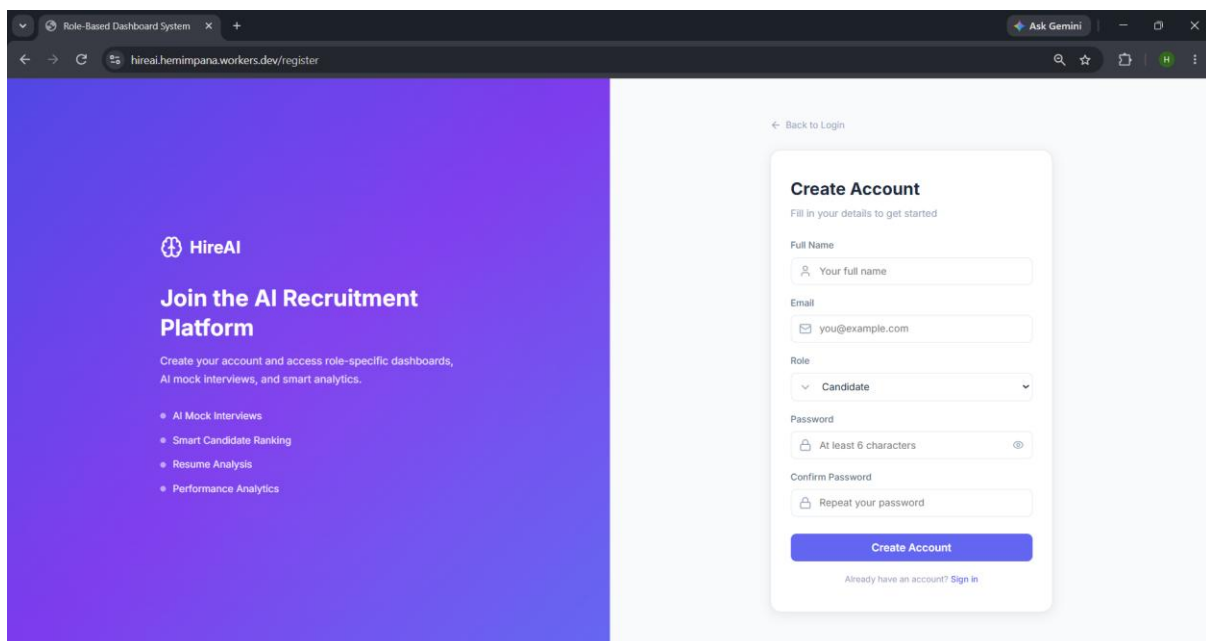


Figure 2. HireAI registration screen.

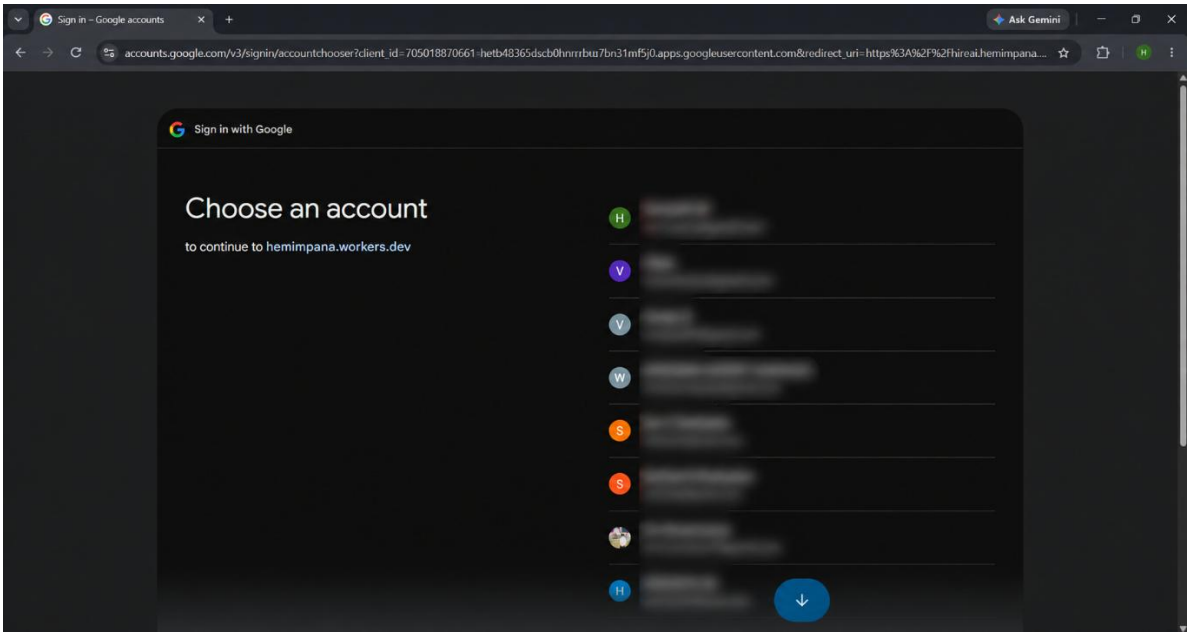


Figure. Third-party authentication options (Google OAuth).

Candidate Experience

Recruitment Platform

Dashboard

Overview

Performance

Reports

Interviews

Past AI Interviews

Upcoming

Mock Interview

AI Feedback

Progress

Profile

Resume

Skills

Settings

Hemant M

Candidate

Logout

Candidate Dashboard

Search...

Welcome back, Hemant!

You have completed 12 AI mock interviews. Keep sharpening your skills!

Start Mock Interview

63/100

Average Score

AI Evaluated

78/100

Highest Score

Personal Best

12

Total Interviews

12 completed

Active

Status

Mock Prep

Quick Actions

Upload Resume

Mock Interview

Past Interviews

Download Report

Upcoming Interviews

No upcoming interviews scheduled.

Past AI Interviews

View All (12)

ROLE	TYPE	DIFFICULTY	SCORE	DURATION	DATE	RECOMMENDATION	ACTIONS
Machine Learning Engineer 3 / 10 questions	Mixed	Medium	16/100	10m 27s	15 Sept 2026	Not Recommended	View Details
Machine Learning Engineer 0 / 10 questions	Mixed	Medium	0/100	0m 35s	15 Sept 2026	Not Recommended	View Details
Machine Learning Engineer 3 / 3 questions	Technical	Easy	78/100	2m 5s	13 Sept 2026	Consider	View Details
Machine Learning Engineer 3 / 3 questions	Technical	Easy	73/100	1m 41s	12 Sept 2026	Consider	View Details
Machine Learning Engineer 0 / 3 questions	Technical	Easy	74/100	0m 50s	1 Sept 2026	Consider	View Details
Machine Learning Engineer 1 / 3 questions	Mixed	Medium	71/100	1m 31s	1 Sept 2026	Not Recommended	View Details

Figure 3. Candidate dashboard overview.

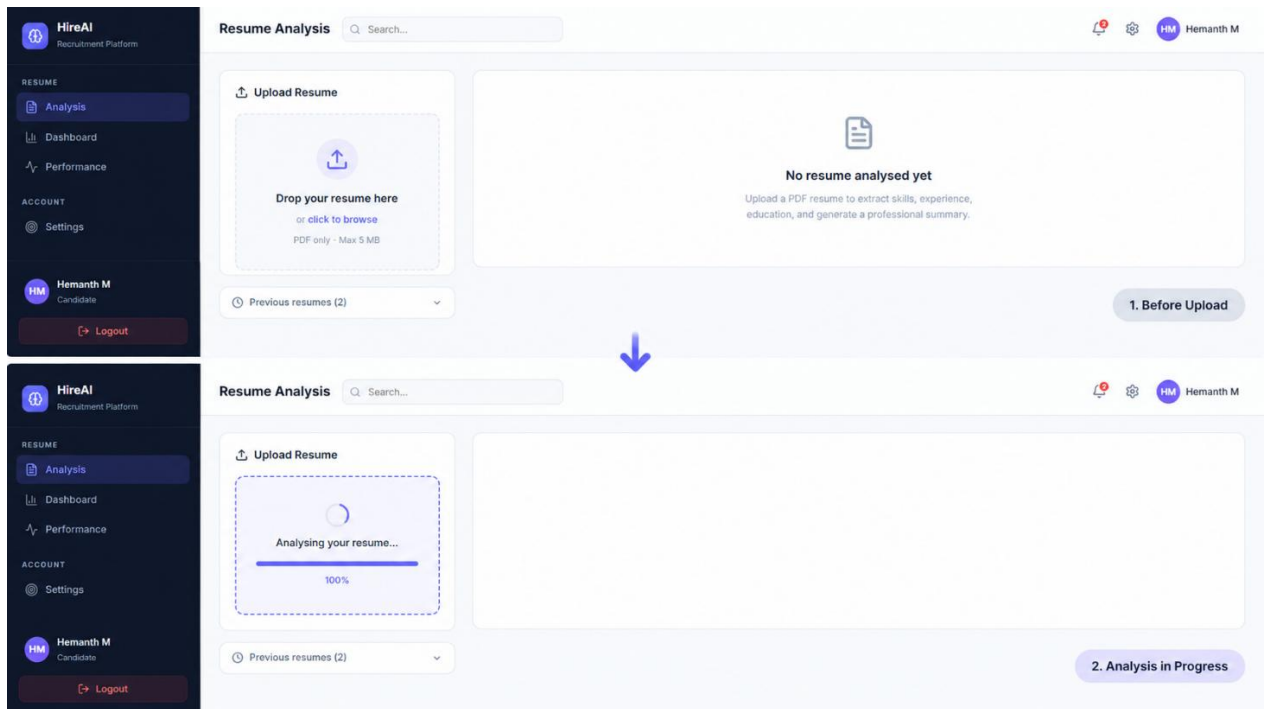


Figure 4. Resume upload and analysis interface.

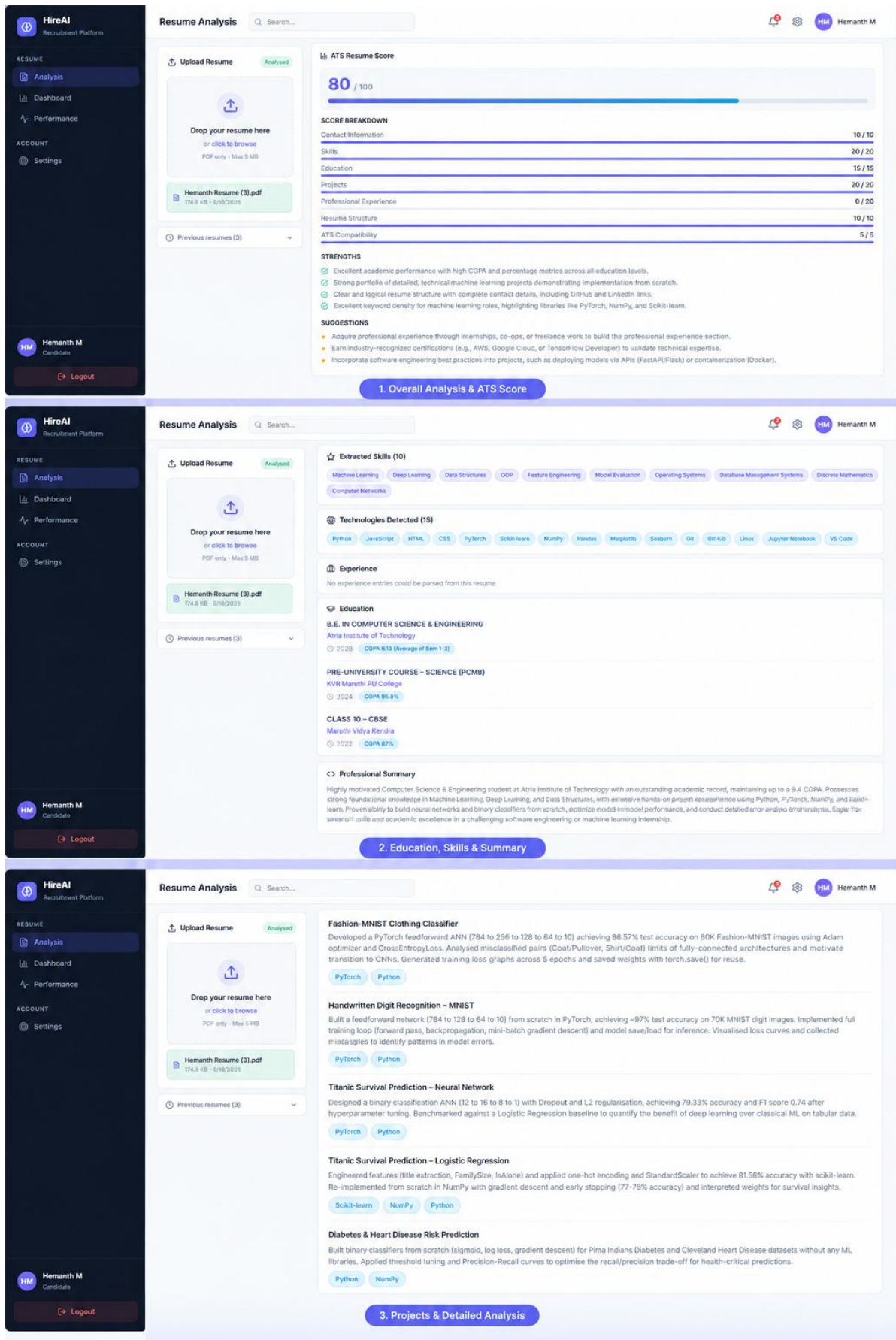


Figure 5. Resume analysis result.

Upload Your Resume

Let AI analyze your resume and suggest the best matching roles.

1 Select Your Resume

Choose a resume to get AI-powered role recommendations, or skip to choose a role manually.

AI Mock Interview

Select a resume to get AI-powered role recommendations, or skip to choose a role manually.

SELECT RESUME (OPTIONAL)

Hemanth Resume (3).pdf
9/16/2026

Hemanth Resume (3).pdf
9/16/2026

Hemanth Resume (3).pdf
9/12/2026

[← Back](#)
[Skip → Manual Role](#)
[Get AI Role Recommendations >](#)

- Secure & Private
- AI-Powered Analysis
- Get Matched Roles
- Start Your Interview

Personalized Role Recommendations

Our AI analyzes your skills, projects, and experience to suggest the best roles for you.

2 Get AI Role Recommendations

Based on your resume analysis, explore the best matching roles or set a custom role.

Recommended Roles

Based on your resume analysis

Machine Learning Engineer 98%

The candidate has built multiple deep learning and machine learning models from scratch using PyTorch, NumPy, and Scikit-learn. Their projects demonstrate strong skills in neural network architecture, hyperparameter tuning, and model evaluation. This aligns perfectly with entry-level Machine Learning Engineer requirements.

Python Developer 95%

Python is the candidate's primary programming language, used extensively to implement complex algorithms, data structures, and machine learning pipelines from scratch. Their deep familiarity with the Python ecosystem, including NumPy and Pandas, makes them highly suitable for this role.

Data Scientist 92%

The candidate possesses strong skills in feature engineering, data visualization using Matplotlib and Seaborn, and statistical model evaluation. Their projects, such as the Titanic survival and healthcare risk predictions, showcase their ability to extract insights and build predictive models from tabular data.

AI Engineer 89%

With hands-on experience developing feedforward artificial neural networks in PyTorch and analyzing classification errors, the candidate has a solid foundation in artificial intelligence. Their academic focus on deep learning and discrete mathematics supports the design and deployment of AI models.

Software Engineer 85%

As a Computer Science & Engineering student with a 9.4 CGPA, the candidate has a strong grasp of core software engineering principles including Data Structures, OOP, Operating Systems, and DBMS. Their ability to write clean, structured code in Python and JavaScript makes them a strong candidate for general software engineering roles.

Or enter a custom role:
e.g. Full Stack Developer, Data Scientist...

INTERVIEW TYPE

DIFFICULTY

NUMBER OF QUESTIONS

Mixed Medium 10

[← Back](#)
[Generate Interview >](#)

- AI-Powered Role Matching
- Detailed Role Insights
- Customize Your Interview
- Choose Difficulty & Type

Realistic Interview Experience

Face AI-powered questions with camera and microphone support for a real interview feel.

3 Start Your AI Mock Interview

Take your interview with AI in a realistic and interactive environment.

AI Mock Interview

Machine Learning Engineer Mixed Medium

10 AI-generated questions tailored for Machine Learning Engineer role.

10 Questions
2 min each

AI Questions
Machine Learning Engineer

Medium
Difficulty

Instant Feedback
Detailed report

- Ensure you are in a quiet environment
- Allow microphone access when prompted by the interview
- Answer each question within 2 minutes
- Do not navigate away during the interview

[← Change Settings](#)
[Start Interview >](#)

- Camera & Microphone Support
- Real-time AI Questions
- Instant Feedback
- Improve Your Skills

Figure 6. Mock interview setup screen.

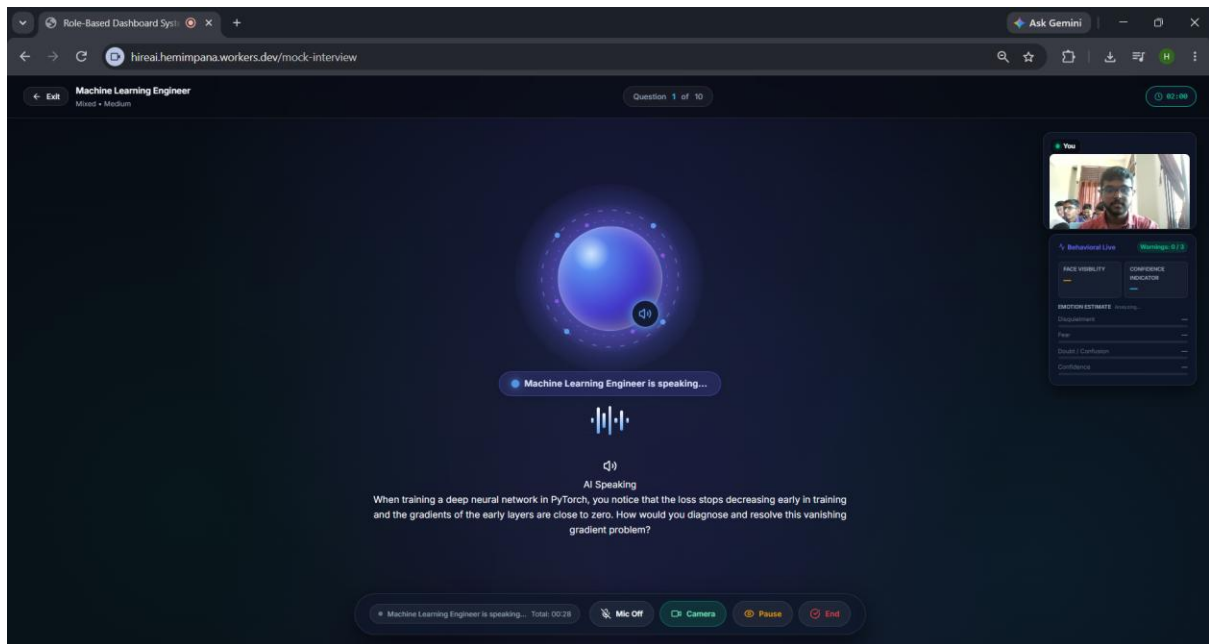


Figure 7. HireAI live mock interview interface.

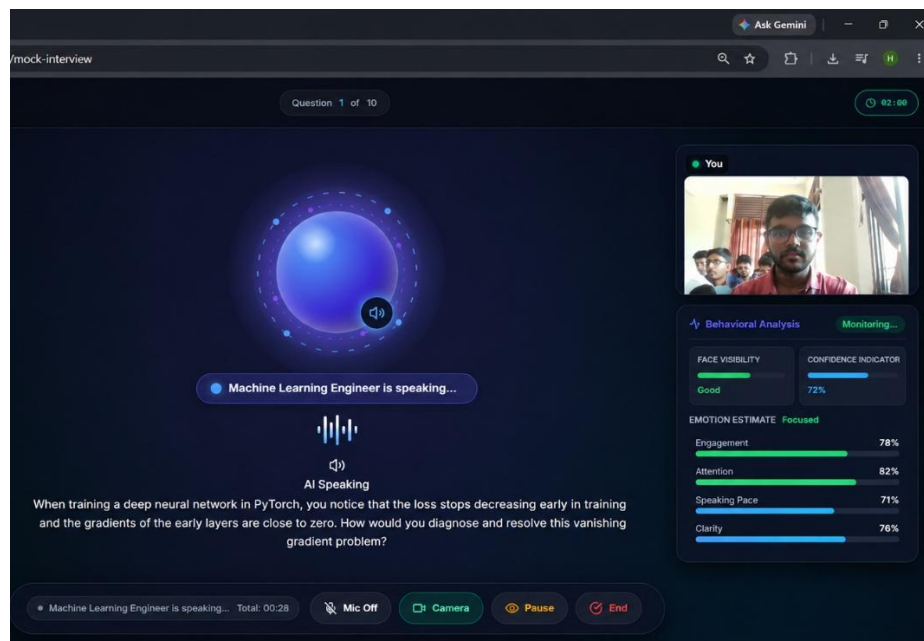


Figure 8. Behavioral analysis panel.

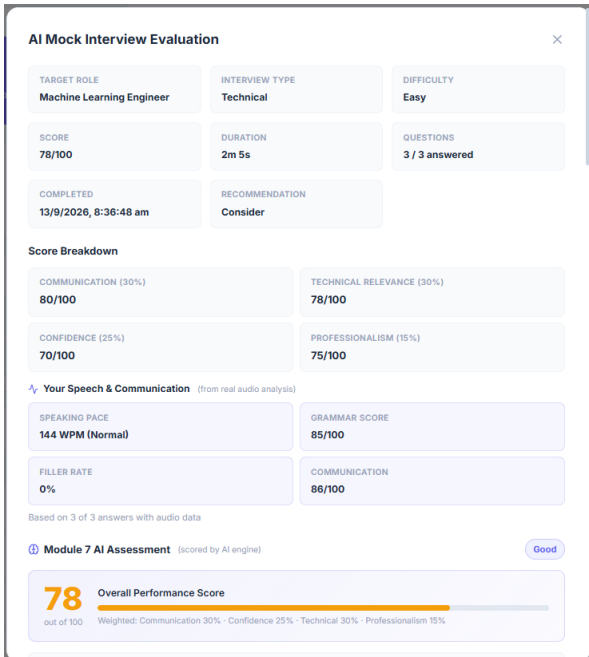


Figure 9. Interview score breakdown.

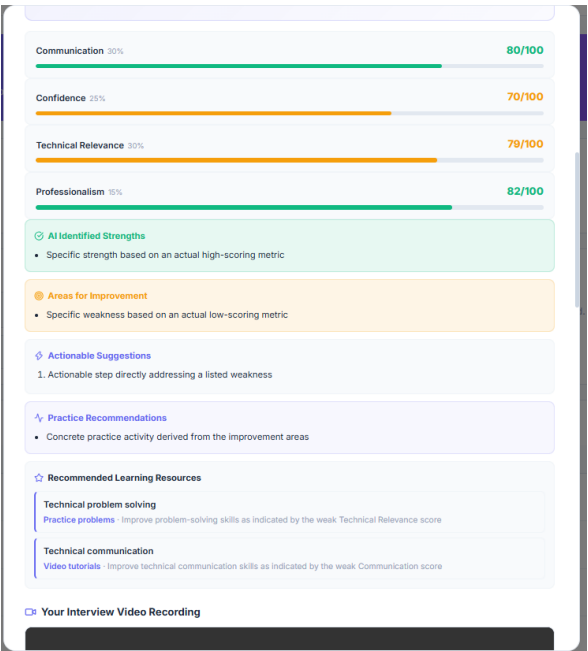


Figure 10. AI-generated feedback.

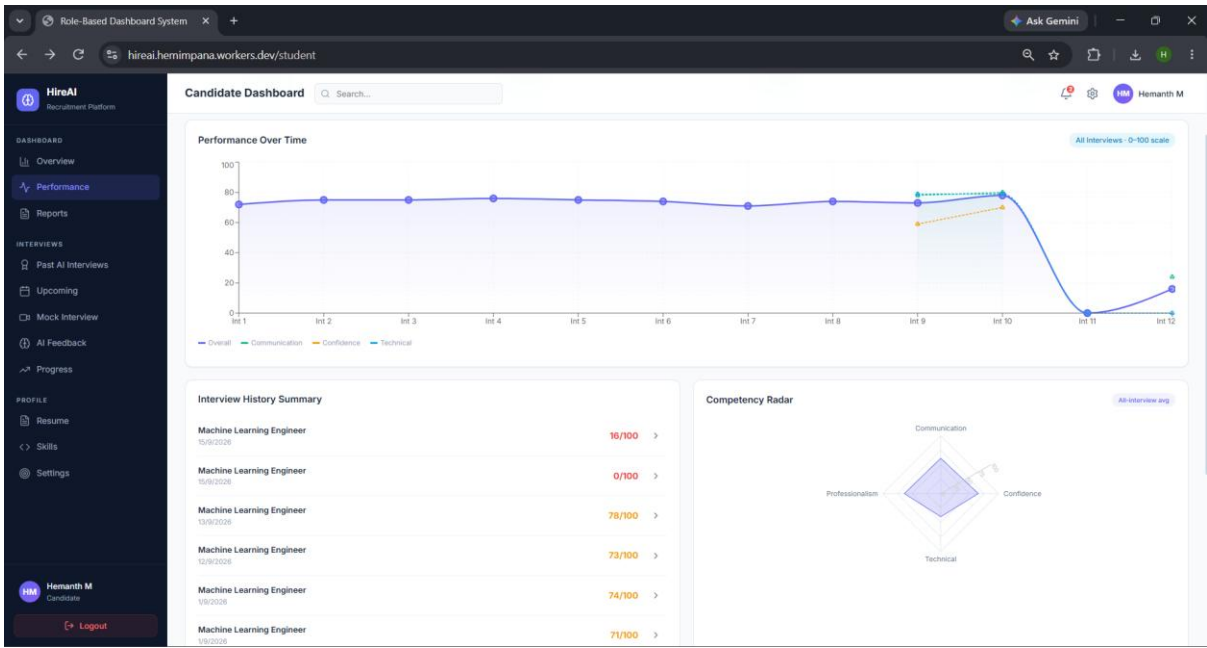


Figure 11. Candidate competency analytics. & Figure 12. Performance trend and weak-area analysis.

Candidate Dashboard Search...

Past AI Mock Interviews
Review all your completed interviews, evaluation breakdowns, and recordings. [Practice Now](#)

ROLE	TYPE	DIFFICULTY	SCORE	DURATION	DATE	RECOMMENDATION	ACTIONS
Machine Learning Engineer 3 / 10 questions	Mixed	Medium	16/100	10m 27s	15 Sept 2026	Not Recommended	View Details
Machine Learning Engineer 6 / 10 questions	Mixed	Medium	0/100	0m 35s	15 Sept 2026	Not Recommended	View Details
Machine Learning Engineer 3 / 3 questions	Technical	Easy	78/100	2m 5s	13 Sept 2026	Consider	View Details
Machine Learning Engineer 3 / 3 questions	Technical	Easy	73/100	1m 41s	12 Sept 2026	Consider	View Details
Machine Learning Engineer 6 / 3 questions	Technical	Easy	74/100	0m 50s	1 Sept 2026	Consider	View Details
Machine Learning Engineer 1 / 3 questions	Mixed	Medium	71/100	1m 31s	1 Sept 2026	Not Recommended	View Details
Machine Learning Engineer 6 / 10 questions	Mixed	Medium	74/100	0m 35s	1 Sept 2026	Consider	View Details
Machine Learning Engineer 6 / 3 questions	Technical	Easy	75/100	0m 36s	1 Sept 2026	Not Recommended	View Details
Machine Learning Engineer 2 / 10 questions	Technical	Medium	76/100	0m 47s	31 Aug 2026	Not Recommended	View Details
Machine Learning Engineer 3 / 3 questions	Mixed	Medium	75/100	1m 27s	14 Aug 2026	Consider	View Details
Machine Learning Engineer 2 / 3 questions	Aptitude	Medium	75/100	1m 23s	14 Aug 2026	Consider	View Details

Figure 13. Candidate interview history.

Recruiter Experience

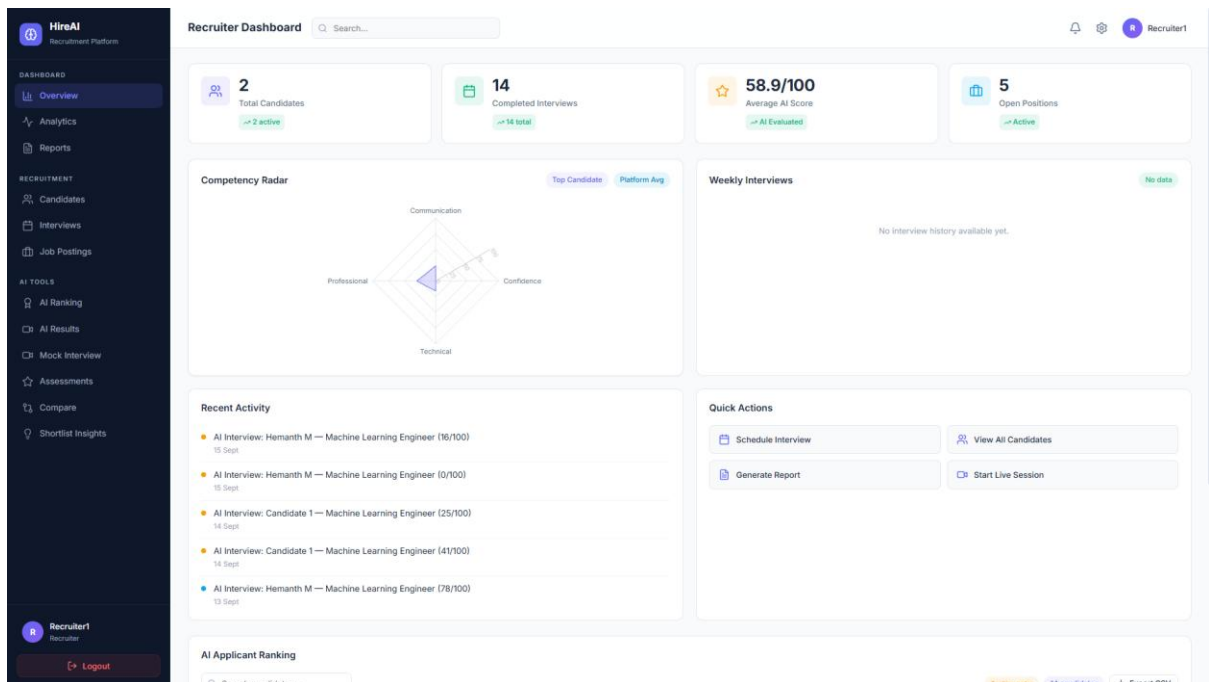


Figure 14. Recruiter dashboard overview.

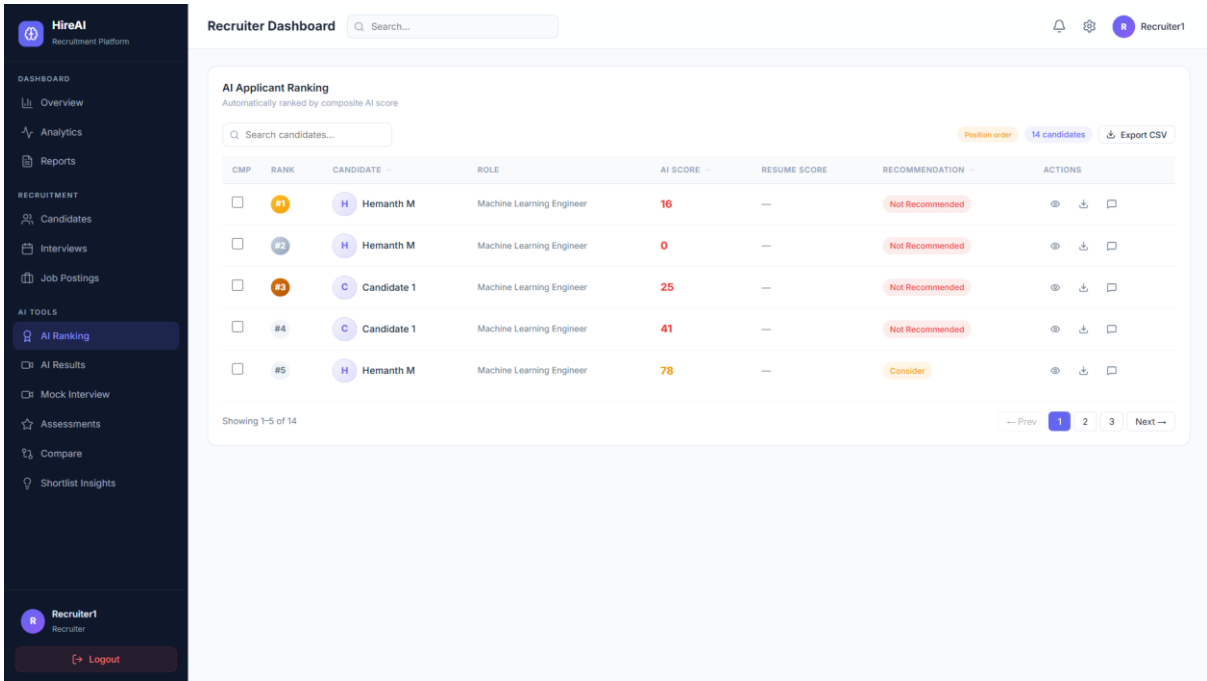
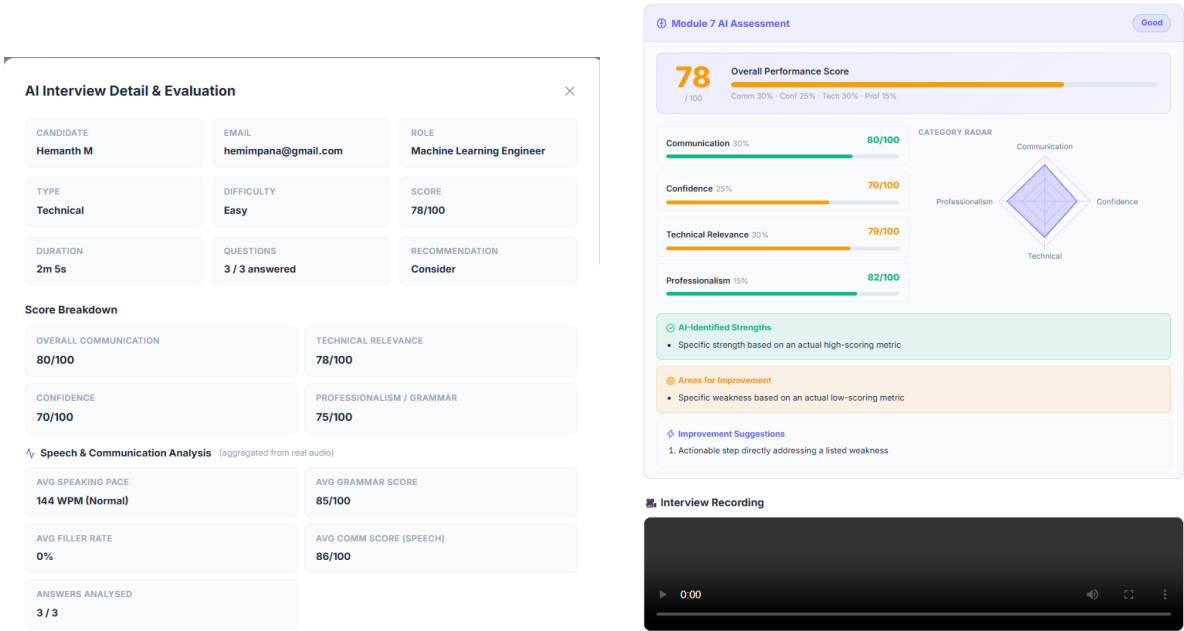


Figure 15. Candidate ranking table.



Candidate Comparison

Side-by-side comparison of 2 selected candidates

Clear Selection
Download

Metric	H HEMANTH M MACHINE LEARNING ENGINEER #6	H HEMANTH M MACHINE LEARNING ENGINEER #7
Overall AI Score	73/100	74/100
Resume / ATS Score	68/100	72/100
Performance Rating	★ 4.0 / 5.0	★ 4.2 / 5.0
Communication (30%)	22/30	<u>24/30</u>
Confidence (25%)	18/25	<u>20/25</u>
Technical (30%)	21/30	<u>22/30</u>
Professionalism (15%)	12/15	<u>13/15</u>
Hire Recommendation	Consider	Consider
Rank	6	7
Interview Date	12 Sept 2026	<u>1 Sept 2026</u>

Underlined values indicate the best result in each row. '-' indicates data not available.

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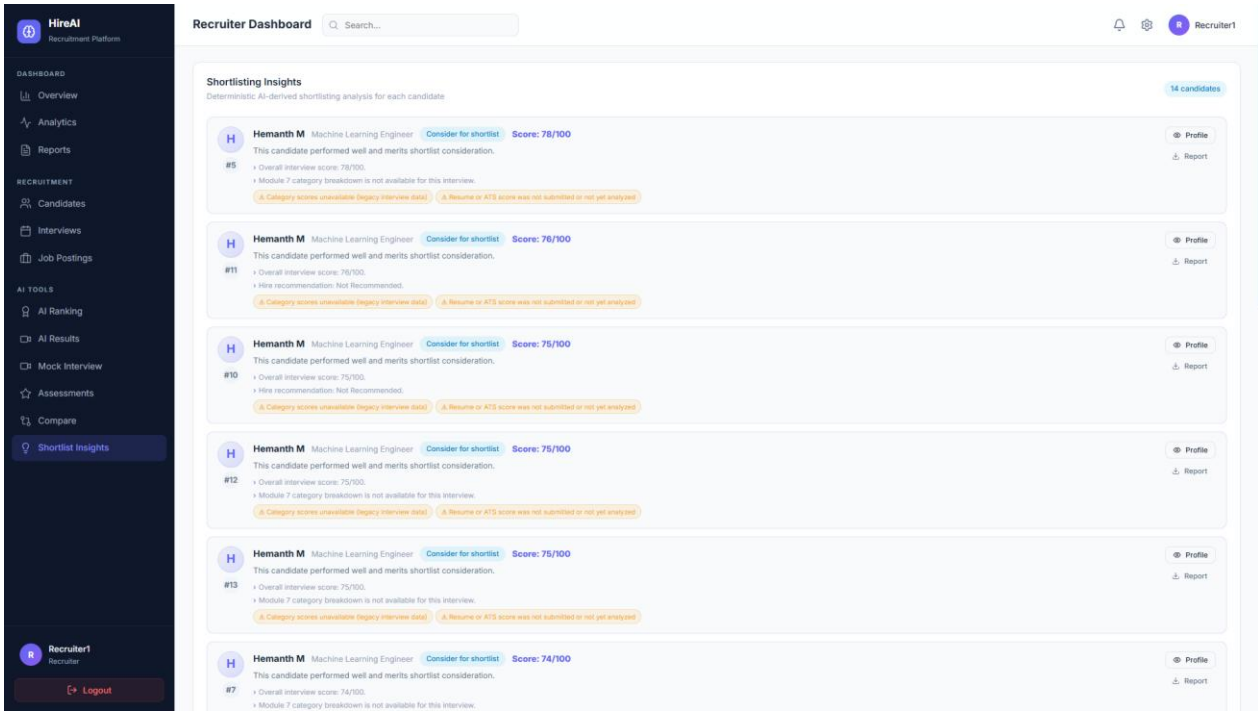


Figure 18. Shortlisting insights.

Administrator Experience

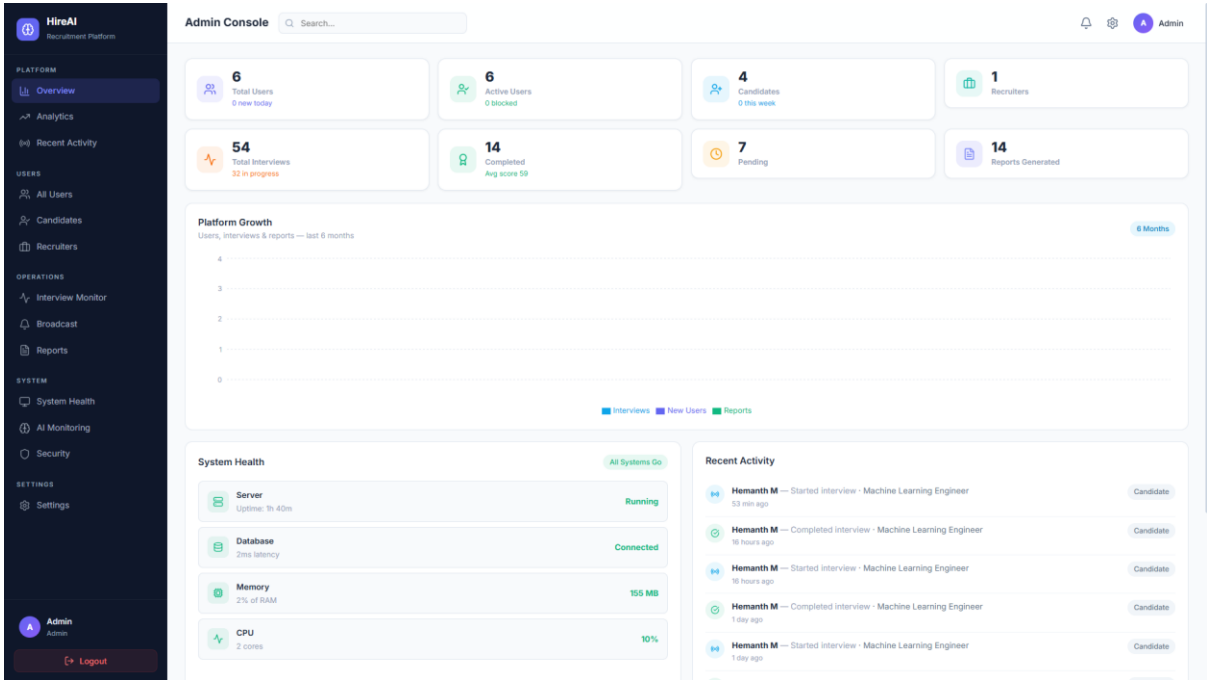


Figure 19. Admin dashboard overview.

SCREENSHOT PLACEHOLDER — FIGURE 20

Admin User Management — Insert: Screenshot showing the user/recruiter management interface, including role or account-status controls.

Figure 20. Admin user management.

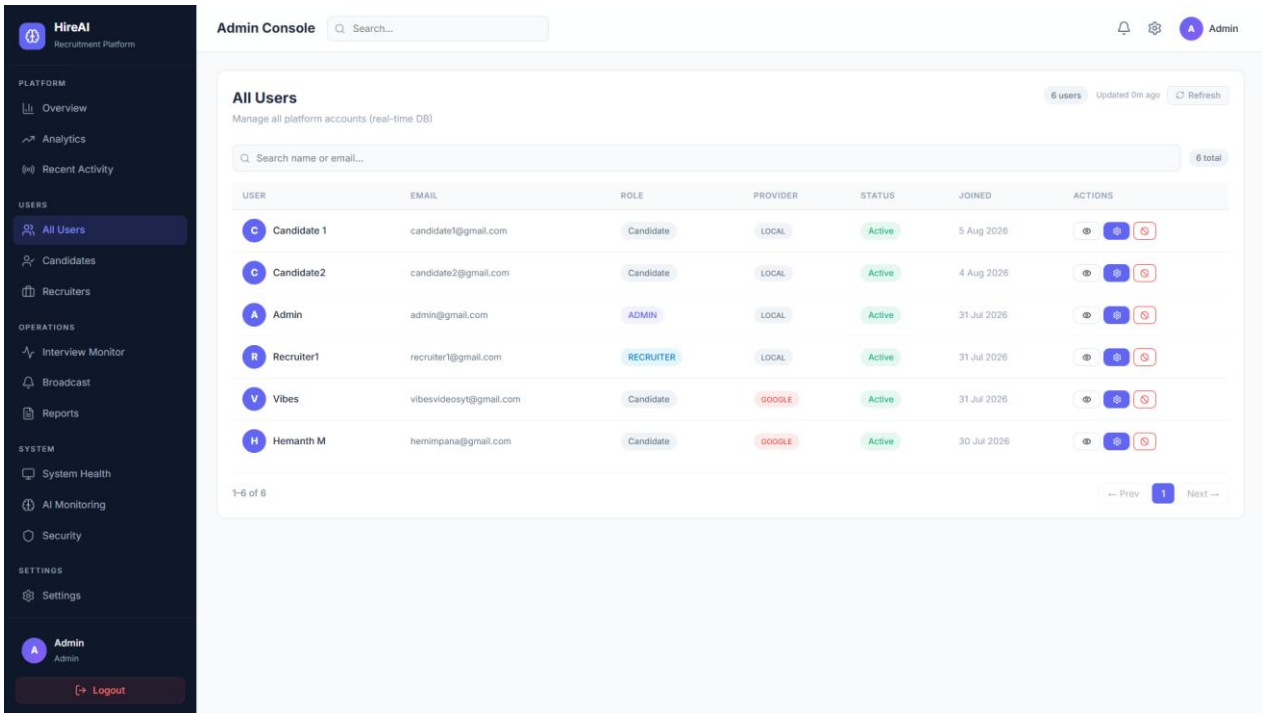


Figure 21. System and AI monitoring.

Account, Scheduling, and Reporting

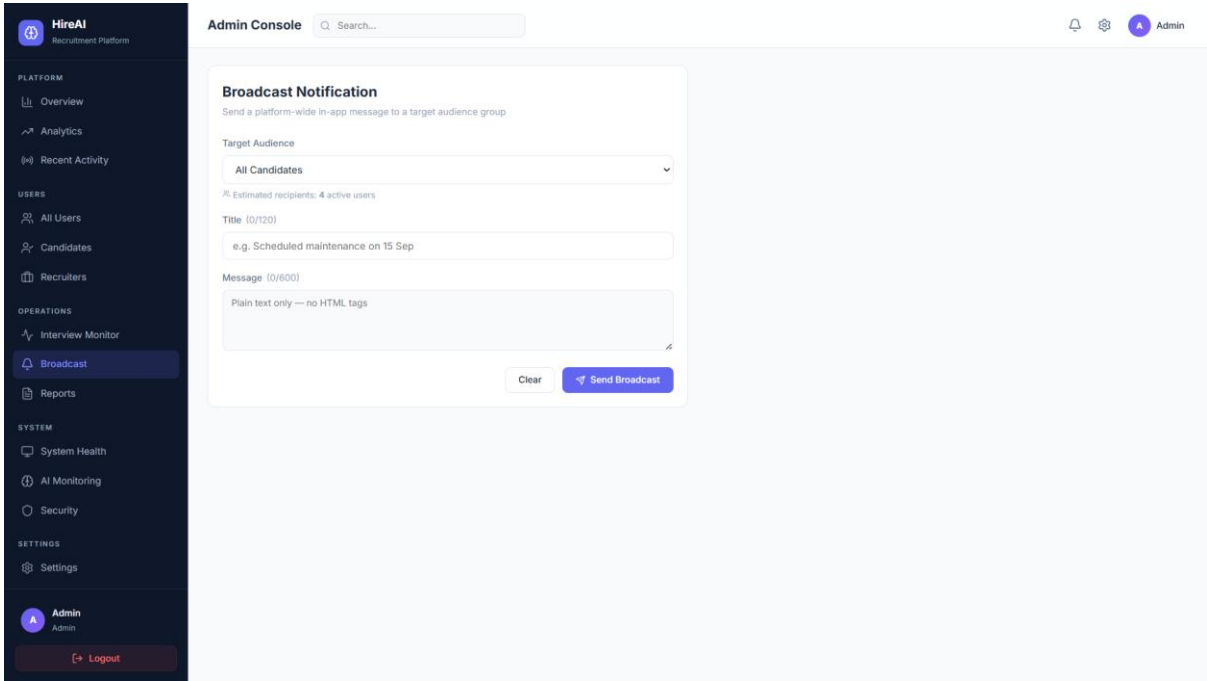


Figure 22. Notifications and scheduling.

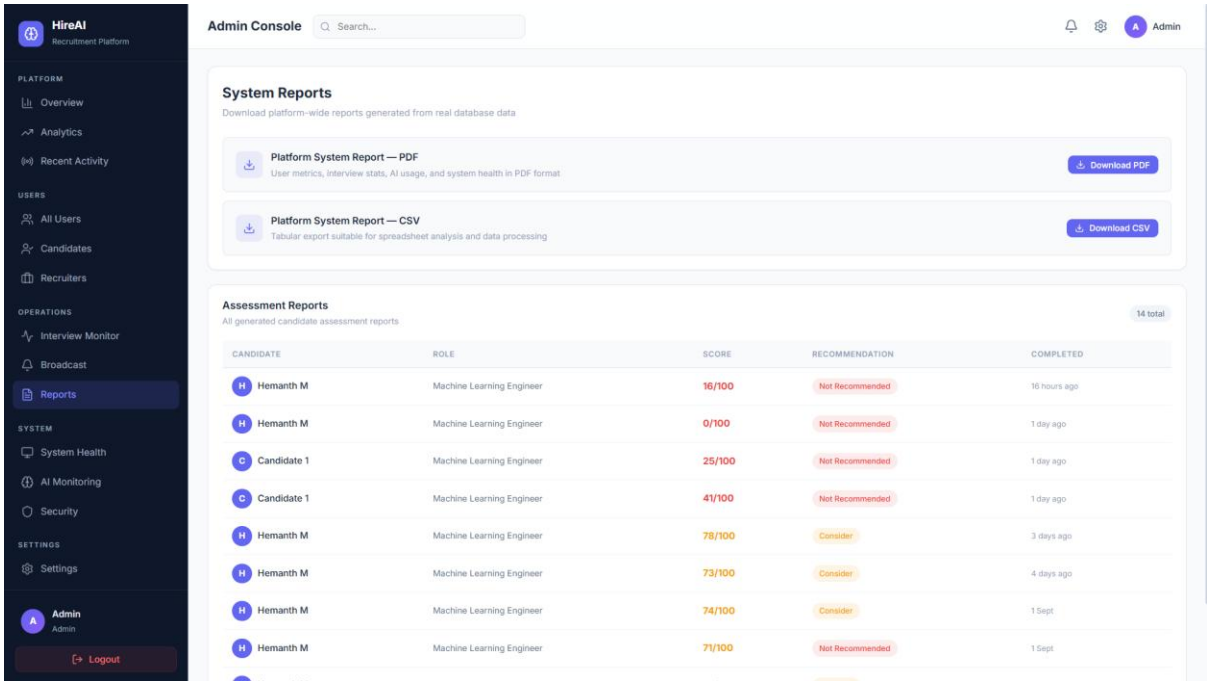


Figure 23. Report and export view.

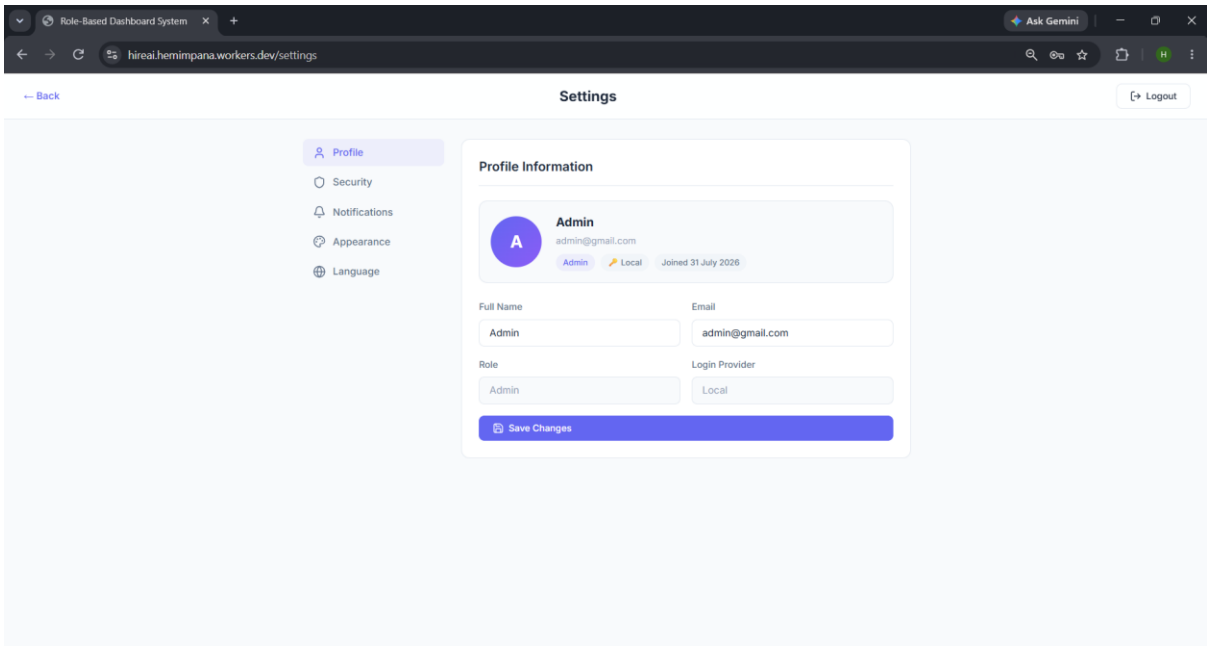


Figure 24. Account settings.

Section 11 — Testing and Validation

11.1 Testing Approach

The project has been reviewed through a series of module-level audits at different points in development, each producing pass/fail assertion counts against a snapshot of the code. Where audit snapshots conflict, this document uses the most recent snapshot for each module: the September 12, 2026 audit as the primary status reference for Module 10, and the September 5, 2026 audit as the reference for Module 7 scoring and feedback. Earlier snapshots, including a September 9, 2026 Module 8 audit that reported fewer implemented requirements, are superseded and are not used as current fact in this document.

Dedicated analytics test files documented in the project include `backend/test_analytics_service.js` and `backend/test_improvement_progress.js`.

11.2 Module 7 Validation

An audit snapshot dated September 5, 2026 reported the following assertion results for Module 7:

Table 11a. Module 7 Test Results (September 5, 2026 audit snapshot)

Test Area	Assertions Passed
Scoring engine	62 / 62
CV rescore	101 / 101
Feedback service	69 / 69
Total	232 / 232 passed, 0 failed

These are audit snapshot values rather than a claim about the currently running code; they should be re-verified if the code has changed since the audit.

11.3 Analytics Validation

A Module 10 audit dated September 12, 2026 reported the following implementation status:

Table 11b. Module 10 Audit Snapshot (12 September 2026)

Requirement Status	Count
Fully implemented	18 / 20
Partially implemented	2 / 20
Missing	0 / 20
Broken	0 / 20

Requirement Status	Count
Mock / static	0 / 20
Unverified	0 / 20

The two partially implemented requirements concern report/profile implementation defects, specifically server-side candidate report data assembly and recruiter candidate-report wiring (see Section 13).

11.4 Build Validation

The September 5, 2026 Module 7 audit also reported that the production frontend build was clean at that time. As with the test results above, this reflects the audited snapshot rather than a standing guarantee for all future builds.

11.5 Known Issues

The current known issues are the two partially implemented Module 10 requirements described above, both relating to server-side report generation and candidate/report data wiring. These are carried forward into the current-limitations list in Section 13.

Section 12 — Deployment

12.1 Local Development

Local development requires a running PostgreSQL instance. Backend dependencies are installed inside backend/, and frontend dependencies are installed at the repository root.

- Backend: cd backend, then npm install, then npm start (or npm run dev for nodemon).
- Frontend: from the repository root, npm install, then npm run dev.
- Default ports: backend 5000, frontend 5173, STT 8765, TTS 8766, CV 8767, and Ollama 11434 when the local LLM provider is used.
- Environment variables are configured in backend/.env and any required frontend environment configuration.

12.2 Codespace Demonstration

The project has been demonstrated running inside a GitHub Codespace, with a Cloudflare Worker used as a gateway/proxy for the public demonstration URL. In the demonstration environment, only the backend (5000) and frontend (5173) ports were intended to be public through the Codespace forwarded-port layer; the database and AI services remained private.

12.3 Current Demo URL

The current demonstration URL is <https://hireai.hemimpana.workers.dev/>. This demonstration architecture depends on the Codespace-hosted application services being alive and reachable, so the demonstration URL should not be treated as an independent, permanent 24/7 production deployment. Codespace processes stop when the Codespace becomes inactive or is stopped, so the project still requires migration to persistent production hosting for continuous, laptop-independent availability.

12.4 Environment Variables

Configuration is loaded from environment variables via dotenv. Categories include database connection settings, the JWT signing secret, Gemini API keys (documented as GEMINI_API_KEY_1 through GEMINI_API_KEY_7), Google and GitHub OAuth client credentials, and email/SMTP settings for Nodemailer. Real values are never reproduced in this document; see Appendix B for the placeholder reference.

12.5 Production Deployment Considerations

The project's deployment documentation also describes the use of process-management tools such as PM2 and reverse proxying for production use, without presenting any single cloud architecture as the only supported production setup. Moving beyond the current development/Codespace

demonstration to persistent hosting, with appropriate resource allocation for the AI services, remains necessary for continuous production availability.

Section 13 — Limitations and Future Work

13.1 Current Limitations

Table 8. Current Limitations

Area	Current Status	Practical Impact
Production deployment	Demonstration runs on GitHub Codespaces behind a Cloudflare Worker gateway.	Not a permanent 24/7 production deployment; availability depends on the Codespace being active.
Local AI resource requirements	STT, TTS, and CV run as separate Python services with their own dependencies.	These features require adequate compute and are not purely browser-based.
Report implementation	Server-side candidate PDF/CSV report assembly and recruiter report wiring.	Latest audit identified implementation defects; treated as a known limitation until repaired.
AI response variability	LLM-generated content can occasionally fail strict JSON expectations or be truncated.	Defensive parsing and fallback handling are required in the pipeline.
AI output interpretation	Scores and AI feedback are decision-support outputs.	Should be interpreted alongside human review rather than as a final decision.
CV service availability	Behavioral analysis depends on the CV service being installed and running.	The behavioral panel and related metrics are unavailable when the CV service is inactive.
Cloud scaling	Current architecture is suited to development and demonstration use.	Production deployment requires persistent hosting and appropriate resource allocation.

13.2 Production Hosting

Migrating from the current Codespace-based demonstration to persistent cloud hosting is the most direct route to a genuinely independent 24/7 deployment.

13.3 AI Service Scaling

The STT, TTS, and CV services each require adequate compute resources; scaling the platform beyond a single demonstration environment would require dedicated hosting or managed infrastructure for these services rather than running them alongside the main application process.

13.4 Reporting Improvements

The server-side candidate PDF/CSV report assembly and recruiter candidate-report wiring defects identified in the Module 10 audit should be corrected so that server-generated reports can be verified and relied upon alongside the existing client-side text report.

13.5 Reliability and Observability

Because LLM-generated content can occasionally fail strict JSON expectations or be truncated, continued attention to defensive parsing and fallback behavior is warranted. Expanding monitoring beyond the current AI usage telemetry would also improve operational visibility.

13.6 Possible Future Enhancements

- Migration to persistent, independently hosted production infrastructure.
- Repair and verification of server-side PDF/CSV report generation.
- Broader automated test coverage across modules beyond the currently documented test files.
- Evaluation of external calendar integration for scheduling, which is not part of the current implementation.

Section 14 — Conclusion

HireAI combines recruitment workflow management, role-based dashboards, resume analysis, AI-assisted mock interviews, structured interview scoring, behavioral analysis, analytics, recruiter decision-support tools, administrative monitoring, scheduling, notifications, and reporting within a single web platform.

Its architecture separates the frontend, backend, database, and AI service layers, allowing each to evolve independently and supporting a future migration from the current development/demonstration infrastructure to persistent cloud deployment.

The strongest engineering aspects of the project, as reviewed, are its centralized scoring logic, PostgreSQL-backed analytics, role-based access control, modular AI services, asynchronous CV processing, structured AI feedback, security middleware, recruiter ranking/comparison/shortlisting tools, admin telemetry, and test-backed scoring and analytics logic.

The known limitations — principally around production hosting and server-side report generation — are documented in Section 13 and represent the clearest next steps for the project's continued development.

Appendix A — Project Structure

Table 7. Project Structure

Path	Responsibility
src/context/	React context providers (application-wide state)
src/services/	Frontend API service functions
src/pages/	Top-level route pages (Login, Register, dashboards, Mock Interview, Settings, etc.)
src/components/	Reusable UI components
backend/server.js	Backend entry point; mounts routes, initializes the database and the reminder scheduler
backend/config/	Database configuration (database.js) and Passport/OAuth configuration (passport.js)
backend/controllers/	Request-handling logic for each route group
backend/routes/	Express route definitions
backend/middleware/	Authentication, RBAC, and request-processing middleware
backend/models/	Data-access logic
backend/services/	Business logic and integration services
backend/ai/	Python AI services (stt_service.py, tts_service.py, cv_service.py) and related model files
package.json, package-lock.json	Frontend dependency manifest
index.html, vite.config.js	Frontend build configuration

Appendix B — Important Configuration Variables

The variables below are configured through environment variables (loaded via dotenv), typically in backend/.env. Values shown are placeholders; no real credentials are reproduced in this document.

Table B-1. Environment Variable Reference

Variable	Purpose	Example Placeholder
Database connection settings	PostgreSQL connection credentials for the ai_recruitment database	<DB_PASSWORD>
JWT_SECRET	Signing key for JSON Web Tokens used in API authentication	<JWT_SECRET>
GEMINI_API_KEY_1 ... GEMINI_API_KEY_7	Gemini cloud LLM API keys used by @google/genai	<GEMINI_API_KEY>
GOOGLE_CLIENT_ID / GOOGLE_CLIENT_SECRET	Google OAuth 2.0 credentials for the Passport Google strategy	<GOOGLE_CLIENT_ID>
GITHUB_CLIENT_ID / GITHUB_CLIENT_SECRET	GitHub OAuth credentials for the Passport GitHub strategy	<GITHUB_CLIENT_ID>
SMTP / email transport settings	Nodemailer configuration for outgoing notification email	<SMTP_PASSWORD>
PORT	Backend server port (default 5000)	5000

Never expose actual API keys, OAuth secrets, passwords, JWT secrets, or database credentials in this or any derived document.

Appendix C — API Category Reference

This appendix restates the API grouping from Section 7.5 as a standalone quick reference.

Table C-1. API Category Quick Reference

Area	Base Path	Purpose
Authentication	/api/auth	Registration, login, profile, password management, logout, Google/GitHub OAuth
Resume	/api/resume	Resume upload, parsing, analysis, status, and result retrieval
Interviews	/api/interviews, /api/interview	Interview creation, question generation, completion, history, and detail retrieval
Recordings	/api/recordings	Recording upload, interview results, and detail retrieval
Speech-to-Text	/api/stt	Speech transcription and speech-related analysis
Computer Vision	/api/cv	Behavioral / video analysis
Analytics	/api/analytics	Candidate and recruiter analytics
Administration	/api/admin	User management and platform telemetry
Notifications	/api/notifications	In-app notification actions
Schedules	/api/schedules	Scheduled interview operations
Reports	/api/reports	Candidate, recruiter, and administrative reporting
Health	/api/health	Service health check

Appendix D — Screenshot Index

Table 9. Screenshot Index

Figure	Screen	What to Capture
Figure 1	Login Page	A full-page screenshot of the HireAI login screen showing the application branding, the login form, and the Google/GitHub authentication options if visible.
Figure 2	Registration Page	The HireAI registration page showing the account creation form.
Figure 3	Candidate Dashboard Overview	A full-width screenshot of the candidate dashboard showing the main summary cards and the primary candidate navigation.
Figure 4	Resume Upload / Resume Analysis	Screenshot of the resume upload/analysis interface immediately after selecting or uploading a PDF, including any visible ATS score, skills, technologies, or analysis result.
Figure 5	Resume Analysis Result	Screenshot showing the parsed resume result, including extracted skills/technologies and ATS-related information.
Figure 6	Role Recommendation / Interview Setup	Screenshot of the mock interview setup page showing the selected role, interview type, difficulty, and question-count controls.
Figure 7	Live Mock Interview	A full-page screenshot of the live mock interview screen showing the current question, timer, microphone/camera controls, and the behavioral-analysis panel.
Figure 8	Behavioral / CV Analysis Panel	Close-up screenshot of the behavioral analysis panel, showing face visibility, confidence indicator, emotion estimate, and engagement-related metrics.
Figure 9	Interview Evaluation / Score Breakdown	Screenshot of the post-interview evaluation view showing the overall score and the four Module 7 categories: Communication, Confidence, Technical Relevance, Professionalism.
Figure 10	AI Feedback	Screenshot showing Strengths, Weaknesses, Suggestions, Practice Recommendations, and Learning Resources.
Figure 11	Candidate Competency Analytics	Screenshot of the candidate analytics tab showing the competency radar chart and category-wise performance.
Figure 12	Performance Trend / Weak Areas	Screenshot showing the candidate's performance trend chart together with the weak-area or improvement-progress section.
Figure 13	Interview History	Screenshot of the candidate's interview history table showing role, score, date, duration, and related actions.

Figure	Screen	What to Capture
Figure 14	Recruiter Dashboard	Full-width screenshot of the recruiter dashboard overview showing major statistics, ranking/activity area, and navigation.
Figure 15	Candidate Ranking	Screenshot of the recruiter ranking table with candidate names, scores, ranking, recommendation, search/sort controls, and pagination if visible.
Figure 16	Candidate Detail Modal	Screenshot of a recruiter opening a candidate profile/detail view containing interview details, scores, and transcript/recording/CV data where available.
Figure 17	Candidate Comparison	Screenshot of the multi-candidate side-by-side comparison screen showing up to four selected candidates and highlighted metrics.
Figure 18	Shortlisting Insights	Screenshot of the recruiter's shortlist-insight area showing tier/category, reasons, strengths, and risks.
Figure 19	Admin Dashboard Overview	Full-page screenshot of the admin dashboard overview.
Figure 20	Admin User Management	Screenshot showing the user/recruiter management interface, including role or account-status controls.
Figure 21	Admin System / AI Monitoring	Screenshot showing the system health, usage, and AI telemetry sections of the admin dashboard.
Figure 22	Notifications / Scheduled Interview	Screenshot of the notifications area and/or scheduled interview/reminder interface.
Figure 23	Report / Export View	Screenshot of the report/export control or a generated report preview/download UI.
Figure 24	Settings / Profile	Screenshot of the account settings page showing profile or preference controls.
Figure 25	OAuth Flow	Screenshot showing the Google/GitHub login options on the login page (not the OAuth provider's private consent page).